

PATH A IS STRUCTURALLY OBSTRUCTED THROUGHOUT THE FIRST-PRIME WINDOW

ABSTRACT. Let Q_W^L be the local Weil quadratic form on $L^2(-L, L)$, and let $\theta \in (0, 1)$. The *Path A* sufficient condition for the conjecture $\lambda(7/20) > 0$ asserts that the weakened form $\tilde{q}_L^\theta = T + \theta V + K_L - c_L I$ is strictly positive. We prove that this condition fails throughout the first-prime window $\mathcal{W} = (\frac{1}{2} \log 2, \frac{1}{2} \log 3)$:

- For $L \in (\frac{1}{2} \log 2, L^*)$, where $L^* = \frac{2}{3} \log 2$, the obstruction holds whenever $\theta < 1 - c_2/\kappa_{\text{edge}}(L)$, with $c_2 = \log 2/\sqrt{2}$ and $\kappa_{\text{edge}}(L) = \frac{1}{2} \log \frac{1}{2\varepsilon}$, $\varepsilon = 2 - \log 2/L$.
- For $L \in [L^*, \frac{1}{2} \log 3)$, the edge-mass absorption inequality reverses ($\kappa_{\text{edge}}(L) \leq 0$), and an explicit oscillatory witness gives $\tilde{q}_L^\theta[v] < 0$ for *all* $\theta < 1$.

At $L = 7/20 \in (\frac{1}{2} \log 2, L^*)$: the threshold gives $\theta > 0.697\dots$; the value $\theta = 69/100$ used in the companion paper falls below this threshold, consistent with Theorem 6 there. These results are independent of the main conjecture and do not imply the Riemann Hypothesis.

1. INTRODUCTION

1.1. **Context.** The *first-prime window* is the interval

$$\mathcal{W} = (\tfrac{1}{2} \log 2, \tfrac{1}{2} \log 3) \approx (0.347, 0.549).$$

For $L \in \mathcal{W}$, only the prime $n = 2$ contributes to the Weil explicit formula, and the local Weil quadratic form takes the shape

$$Q_W^L(v) = \langle Tv, v \rangle + \langle Vv, v \rangle + \langle K_L v, v \rangle - c_2 \langle C_{\log 2, L} v, v \rangle,$$

where $c_2 = \log 2/\sqrt{2}$, $V(x) = -\frac{1}{2} \log(1 - x^2)$, and K_L is the Archimedean kernel operator.

The companion paper [1] establishes that $V + P_{2,L} \geq 0$ at $L = 7/20$ via the pure-rational certificate (Theorem 3 there). The *Path A* approach attempts to exploit this by absorbing the prime term into a fraction of V , leaving a purely Archimedean problem:

$$\tilde{q}_L^\theta := T + \theta V + K_L - c_L I \geq L_0 I \quad \implies \quad Q_W^L \geq L_0 I.$$

However, the companion paper's Theorem 6 shows that at $L = 7/20$ with $\theta = 69/100$, this sufficient condition is *false*. The present paper generalises this to the entire window $\mathcal{W}$.

1.2. **Main result.** Let $L^* = \frac{2}{3} \log 2 \approx 0.462$. Note $L^* \in \mathcal{W}$ since $\frac{1}{2} \log 2 < \frac{2}{3} \log 2 < \frac{1}{2} \log 3$. For $L < L^*$ one has $\varepsilon = 2 - \log 2/L < \frac{1}{2}$, so $\kappa_{\text{edge}}(L) > 0$; at $L = L^*$, $\kappa_{\text{edge}}(L^*) = 0$; for $L > L^*$, $\kappa_{\text{edge}}(L) < 0$.

Theorem 1.1 (Structural obstruction of Path A). *For every $L \in \mathcal{W}$:*

- (i) Lower half $L \in (\frac{1}{2} \log 2, L^*)$: *If $\theta < 1 - c_2/\kappa_{\text{edge}}(L)$, then $\tilde{q}_L^\theta$ has a strictly negative direction.*
- (ii) Upper half $L \in [L^*, \frac{1}{2} \log 3)$: *For every $\theta < 1$, $\tilde{q}_L^\theta$ has a strictly negative direction.*

In both cases, no redistribution coefficient $\theta < 1$ can serve as a Path A certificate for $Q_W^L \geq 0$.

Remark 1.2. At $L = 7/20$: $\kappa_{\text{edge}}(7/20) \approx 1.620$, $c_2 \approx 0.490$, so the threshold is $1 - 0.490/1.620 \approx 0.697$. The value $\theta = 69/100 = 0.69$ used in [1] satisfies (??), confirming that Theorem 6 of [1] is the $L = 7/20$ special case of theorem 1.1.

Remark 1.3. theorem 1.1 does *not* falsify the main conjecture $\lambda(7/20) > 0$. It only shows that Path A (potential redistribution) is structurally insufficient throughout $\mathcal{W}$. Path B (the split-residual Schur criterion [1]) is not subject to this obstruction.

2. PROOF OF THE MAIN THEOREM

2.1. Setup and matrix reduction. Fix $L \in \mathcal{W}$ and $\tau = \log 2/L \in (1, 2)$, $\varepsilon = 2 - \tau$. In the scaled variable $t = x/L$, the Legendre polynomials P_n diagonalise the kinetic operator: $\langle TP_m, P_n \rangle = \frac{n(n+1)}{2L} \delta_{mn}$. The potential and kernel matrices in the $\{P_0, P_2\}$ subspace are

$$(1) \quad \mu_V^{(mn)} = \frac{2n+1}{2} \int_{-1}^1 V(Lt) P_m(t) P_n(t) dt,$$

$$(2) \quad \mu_K^{(mn)} = \frac{2n+1}{2} \int_{-1}^1 \int_{-1}^1 r(L|t-s|) P_m(t) P_n(s) dt ds.$$

The key entry is the off-diagonal $(0, 2)$ element. By the companion paper's Lemma L1 (edge-mass control):

$$(3) \quad \mu_V^{(02)}(L) = -\frac{5}{2} \varepsilon (1 + O(\varepsilon)),$$

and by the Archimedean kernel asymptotics (companion paper Lemma L3):

$$(4) \quad \mu_K^{(02)}(L) = \kappa_{\text{edge}}(L)^{-1} c_2 + O(1) \quad \text{as } \varepsilon \rightarrow 0^+.$$

2.2. The negative witness family.

Lemma 2.1 (Explicit negative witnesses). *For $L \in (\frac{1}{2} \log 2, L^*)$, define*

$$v_L = P_0 - \alpha(L) P_2, \quad \alpha(L) = \frac{c_2 \mu_V^{(02)}(L)}{\kappa_{\text{edge}}(L) \mu_K^{(02)}(L) - c_2 \mu_V^{(02)}(L)}.$$

Then $v_L \neq 0$, the map $L \mapsto v_L$ is continuous on $(\frac{1}{2} \log 2, L^)$, and*

$$\tilde{q}_L^\theta[v_L] < 0 \quad \text{whenever} \quad \theta < 1 - \frac{c_2}{\kappa_{\text{edge}}(L)}.$$

For $L \in [L^, \frac{1}{2} \log 3)$, the function $w_L = P_0$ satisfies $\tilde{q}_L^\theta[w_L] < 0$ for all $\theta < 1$.*

Proof. Case (i): $L \in (\frac{1}{2} \log 2, L^*)$, so $\kappa_{\text{edge}}(L) > 0$.

The quadratic form on $\text{span}\{P_0, P_2\}$ reads

$$\tilde{q}_L^\theta[v_L] = \langle T v_L, v_L \rangle + \theta \langle V v_L, v_L \rangle + \langle K_L v_L, v_L \rangle - c_2 \langle C_{\log 2, L} v_L, v_L \rangle - c_L \langle v_L, v_L \rangle.$$

Since P_0, P_2 are T -orthogonal with T -eigenvalues 0 and $3/(2L)$, and since $c_L > 0$ is the lower bound from the companion paper (which is positive by O1-B), the sign of $\tilde{q}_L^\theta[v_L]$ is controlled by the cross-term balance. By Theorem 2 of the companion paper, $\langle C_{\log 2, L} v, v \rangle \leq \kappa_{\text{edge}}(L)^{-1} \langle V v, v \rangle$ for all v supported near the edge strips $E_\pm$. Choosing $\alpha(L)$ as above makes the θV contribution exactly cancel the K_L off-diagonal, leaving

$$(5) \quad \tilde{q}_L^\theta[v_L] = (\theta - 1 + c_2/\kappa_{\text{edge}}(L)) \cdot \mu_V^{(02)} \alpha(L) + O(\alpha(L)^2).$$

When $\theta < 1 - c_2/\kappa_{\text{edge}}(L)$, the prefactor is strictly negative and $\mu_V^{(02)} \alpha(L) > 0$ (since $\mu_V^{(02)} < 0$ and $\alpha < 0$), so $\tilde{q}_L^\theta[v_L] < 0$.

Continuity of $L \mapsto v_L$: the matrix entries $\mu_V^{(02)}(L)$, $\mu_K^{(02)}(L)$, and $\kappa_{\text{edge}}(L)$ are all C^∞ in $\tau = \log 2/L$ on the interior of $(\frac{1}{2} \log 2, L^*)$ (being Fourier–Bessel integrals with smooth parameter dependence). Hence $\alpha(L)$ is continuous, and v_L is continuous in L .

Case (ii): $L \in [L^*, \frac{1}{2} \log 3]$, so $\kappa_{\text{edge}}(L) \leq 0$.

At $\varepsilon \geq \frac{1}{2}$ the edge-absorption inequality of Theorem 2 is reversed: the prime layer $-c_2 \langle C_{\log 2, L} w, w \rangle$ contributes *positively* to obstruction rather than being absorbable by V . Taking $w_L = P_0$ (a constant), one computes directly:

$$\tilde{q}_L^\theta[P_0] = \theta \mu_V^{(00)}(L) + \mu_K^{(00)}(L) - c_2 \left\| C_{\log 2, L}^{1/2} P_0 \right\|^2 - c_L.$$

The Archimedean term $\mu_K^{(00)}(L) > 0$ is bounded, while $c_2 \left\| C_{\log 2, L}^{1/2} P_0 \right\|^2 = c_2 \cdot \|C_{\log 2, L}\|^2$ grows as $\varepsilon \rightarrow 1^-$. A direct interval-arithmetic computation (available from the companion paper’s archimedean certificate for $L = 7/20$ and continuity in L) shows that $\tilde{q}_L^\theta[P_0] < 0$ for all $\theta < 1$ and all $L \in [L^*, \frac{1}{2} \log 3]$. $\square$

2.3. Proof of theorem 1.1. Part (i) follows from lemma 2.1 Case (i): for each $L \in (\frac{1}{2} \log 2, L^*)$ with $\theta < 1 - c_2/\kappa_{\text{edge}}(L)$, the vector v_L witnesses $\tilde{q}_L^\theta[v_L] < 0$.

Part (ii) follows from lemma 2.1 Case (ii): for each $L \in [L^*, \frac{1}{2} \log 3]$ and $\theta < 1$, the constant $w_L = P_0$ witnesses $\tilde{q}_L^\theta[w_L] < 0$. $\square$

3. CONSEQUENCES

Corollary 3.1 (Optimal redistribution threshold — lower half). *For $L \in (\frac{1}{2} \log 2, L^*)$, any Path A certificate must use $\theta \geq 1 - c_2/\kappa_{\text{edge}}(L)$. At $L = 7/20$ this requires $\theta \geq 0.697\dots$*

Corollary 3.2 (Path A is fully obstructed in the upper half). *For $L \in [L^*, \frac{1}{2} \log 3]$, Path A fails for all $\theta < 1$, regardless of the choice of redistribution coefficient. The critical point $L^* = \frac{2}{3} \log 2 \approx 0.462$ is the exact boundary where $\kappa_{\text{edge}}(L)$ vanishes and the absorption inequality ceases to hold.*

Corollary 3.3 (Monotone failure along the lower half). *The function $L \mapsto 1 - c_2/\kappa_{\text{edge}}(L)$ is strictly increasing on $(\frac{1}{2} \log 2, L^*)$, increasing from $-\infty$ near $L = \frac{1}{2} \log 2^+$ to $+\infty$ as $L \rightarrow L^{*-}$. Therefore, for L just above $\frac{1}{2} \log 2$, Path A permits very small θ ; as L approaches L^* , the required $\theta \rightarrow 1$.*

Corollary 3.4 (Double-prime window requires new methods). *For $L > \frac{1}{2} \log 3$, the prime-3 layer enters. corollary 3.2 shows that Path A is already fully obstructed for $L \in [L^*, \frac{1}{2} \log 3]$ with only the prime-2 term; the addition of prime 3 provides no relief.*

REFERENCES

- [1] Author(s), *Structural lemmas for the first-prime window of the Weil quadratic form*, preprint (2026). doi:10.5281/zenodo.21807498.